IITGN PG Diploma in AI-ML & Agentic AI Engineering · Cohort 2

Week 2 Practice Paper 1

pandas · SQL · inferential statistics · EDA

A practice paper modelled on the Week 1 Surprise Test. It is not an official course paper.

Rules

- About 50 minutes, 36 items in 6 parts. Pen and paper only: no calculator, laptop or notes. Every number can be worked by hand.
- One mark per item. No part marks and nothing taken off for a wrong answer, so answer every item.
- Starred items (*) need every correct option and no other.
- On a "write the number" item, give the number to the places asked. A wrong rounding is marked wrong.
- Code is Python 3 with pandas, read exactly as written. SQL is standard PostgreSQL.
- Kalpa Retail and everyone in it are fictional. Money is in rupees (Rs); lakh is used for large numbers.

Blueprint

Part	Area	Items	Minutes	Easy	Medium	Hard
1	pandas: groupby and dates	Q1 to Q6	8	1	3	2
2	pandas: joins, loc and apply	Q7 to Q12	9	0	3	3
3	SQL: joins and GROUP BY	Q13 to Q18	8	1	3	2
4	SQL: window functions and CTEs	Q19 to Q23	7	0	3	2
5	Inference: sampling, CI, z-tests, p-values	Q24 to Q31	10	1	5	2
6	EDA: from a table to a finding	Q32 to Q36	7	1	3	1
	Total	**36**	**49**	**4**	**20**	**12**

Step one: rate yourself before you start

1 = never used it · 2 = can follow it when shown · 3 = can do it alone on a small problem · 4 = can find and fix mistakes in someone else's version

Part	Area	Rating (1-4)	Score after marking
1	pandas: groupby and dates		___ / 6
2	pandas: joins, loc and apply		___ / 6
3	SQL: joins and GROUP BY		___ / 6
4	SQL: window functions and CTEs		___ / 5
5	Inference: sampling, CI, z-tests, p-values		___ / 8
6	EDA: from a table to a finding		___ / 5

Which stores are actually busy, and which just look busy?

Kavya Nair, senior analyst at Kalpa Retail, is working on a small export of orders. Two orders lost their amount in a payment-gateway glitch. Assume pandas is imported as `pd`.

Exhibit 1A

```
orders = pd.DataFrame({
    'store':    ['Pune', 'Pune', 'Surat', 'Surat', 'Surat', 'Indore'],
    'order_id': [101, 102, 103, 104, 105, 106],
    'amount':   [500, None, 300, 300, None, 800],
    'customer': ['C1', 'C2', 'C3', 'C3', 'C4', 'C5']
})

r = orders.groupby('store').agg(
    n_rows = ('order_id', 'size'),
    n_amt  = ('amount',   'count'),
    n_cust = ('customer', 'nunique')
)
print(r.loc['Surat'].tolist())
```

Q1 Predict the output · Easy

What does the last line of Exhibit 1A print?

A. `[3, 3, 2]`

B. `[3, 2, 2]`

C. `[3, 2, 3]`

D. `[2, 2, 2]`

Exhibit 1B

```
sales = pd.DataFrame({
    'store':  ['Pune', 'Pune', 'Surat', 'Surat', 'Surat'],
    'amount': [200, 600, 100, 100, 200]
})
```

Q2 Choose the line · Medium

Kavya wants each order's share of its own store's total as a new column, giving `[0.25, 0.75, 0.25, 0.25, 0.5]`. Which line does it?

A. `sales['share'] = sales['amount'] / sales.groupby('store')['amount'].agg('sum')`

B. `sales['share'] = sales['amount'] / sales.groupby('store')['amount'].transform('sum')`

C. `sales['share'] = sales['amount'] / sales['amount'].sum()`

D. `sales['share'] = sales.groupby('store')['amount'].sum() / sales['amount']`

Exhibit 1C

```
s = pd.Series(['03/04/2026', '15/04/2026', '31/04/2026', '01/05/2026'])
d = pd.to_datetime(s, format='%d/%m/%Y', errors='coerce')
print(d.isna().sum(), d.dt.month.nunique())
```

Q3 Predict the output · Medium

The Surat staff typed delivery dates by hand, day first. What does Exhibit 1C print?

A. `0 2`

B. `1 2`

C. `1 3`

D. Nothing: it raises `ValueError` on `'31/04/2026'`

Exhibit 1D

```
daily = pd.DataFrame({
    'store':   ['Pune', 'Pune', 'Surat', 'Surat'],
    'day':     [1, 2, 1, 2],
    'orders':  [10, 30, 20, 20],
    'revenue': [5000, 6000, 6000, 4000]
})
daily['aov'] = daily['revenue'] / daily['orders']
print(daily.groupby('store')['aov'].mean())

# Output:
# store
# Pune     350.0
# Surat    250.0
```

Q4 Write the number · Medium

Kavya puts Pune's average order value on her slide as Rs 350. What is Pune's actual average order value over the two days (total revenue divided by total orders), in rupees?

Answer: _____

Q5* Choose every correct option · Hard

Use the `orders` DataFrame from Exhibit 1A. Which statements are true? Mark every correct option.

- **A.** `orders.groupby('store').filter(lambda g: len(g) >= 2)` returns 5 rows.
- **B.** `orders.groupby('store')['amount'].sum()` gives NaN for Surat, because one of its amounts is missing.
- **C.** `orders.groupby('store', as_index=False)['amount'].sum()` returns a DataFrame with 3 rows and a `store` column.
- **D.** `orders.groupby('store')['amount'].transform('count')` returns 3 values, one per store.
- **E.** `orders.groupby('store')['amount'].mean()` gives 250 for Pune.

** Mark every correct option. Right only with every correct box and no other.*

Exhibit 1E

```
AS_OF = pd.Timestamp('2026-10-01')
cust = pd.DataFrame({
    'cust': ['C1', 'C2', 'C3', 'C4'],
    'last_order': pd.to_datetime(['2026-09-01', '2026-07-03',
                                  '2026-07-02', '2026-06-15'])
})
cust['days_since'] = (AS_OF - cust['last_order']).dt.days
lapsed = cust[cust['days_since'] > 90]
print(len(lapsed))
```

Q6 Predict the output · Hard

Kalpa calls a customer lapsed after more than 90 days without an order. What does Exhibit 1E print?

- **A.** 1
- **B.** 2
- **C.** 3
- **D.** It depends on the day the code is run

Did the join quietly change the revenue?

Kavya joins the order file to the customer master to get each order's city. Anand Iyer, the finance controller, says the joined revenue no longer matches what the bank settled.

Exhibit 2A

```
orders = pd.DataFrame({
    'order_id': [1, 2, 3, 4],
    'cust_id':  ['C1', 'C2', 'C2', 'C9'],
    'amount':   [100, 200, 300, 400]
})
customers = pd.DataFrame({            # the customer master
    'cust_id': ['C1', 'C2', 'C2', 'C3'],
    'city':    [None, 'Surat', 'Surat ', 'Indore']   # note: 'Surat ' has a trailing space
})
```

Q7 Predict the output · Medium

`m = orders.merge(customers, on='cust_id', how='left')` then `print(len(m), m['amount'].sum())`. What is printed?

A. `4 1000`

B. `6 1500`

C. `5 1100`

D. `6 1000`

Q8 Predict the counts · Hard

Kavya deduplicates the master before joining:

```
c1 = customers.drop_duplicates()
c2 = customers.drop_duplicates(subset='cust_id')
print(len(c1), len(c2), len(orders.merge(c2, on='cust_id', how='left')))
```

What is printed?

A. `3 3 4`

B. `4 3 4`

C. `4 3 6`

D. `3 3 6`

Q9 Choose the line · Medium

Anand wants the join to fail loudly whenever the master has a repeated customer. Which `validate=` value states the true relationship of `orders.merge(customers, on='cust_id', how='left', validate=...)`, so that it passes on a clean master (one row per customer) and stops the run on Exhibit 2A?

A. `'one_to_one'`

B. `'one_to_many'`

C. `'many_to_one'`

D. `'many_to_many'`

Q10* Choose every correct option · Hard

Using the original `orders` and `customers` from Exhibit 2A, which lines keep only order 4's row, the order whose customer is missing from the master? Mark every correct option.

A. `orders[~orders['cust_id'].isin(customers['cust_id'])]`

B. `m = orders.merge(customers, on='cust_id', how='left', indicator=True)`, then `m[m['_merge'] == 'left_only']`

C. `orders.merge(customers, on='cust_id', how='inner')`

D. `m = orders.merge(customers, on='cust_id', how='left')`, then `m[m['city'].isna()]`

E. `m = orders.merge(customers, on='cust_id', how='outer', indicator=True)`, then `m[m['_merge'] == 'right_only']`

** Mark every correct option. Right only with every correct box and no other.*

Exhibit 2B

```
df = pd.DataFrame({
    'store':   ['Pune', 'Surat', 'Pune', 'Indore', 'Surat'],
    'amount':  [500, None, 1200, 800, 300],
    'channel': ['app', 'web', 'app', 'app', None]
})
mask = (df['amount'] > 400) & ~(df['channel'] == 'web')
print(df.loc[mask, 'store'].tolist())

def tier(row):
    if row['amount'] >= 500:
        return 'Gold'
    if row['amount'] >= 1000:
        return 'Platinum'
    return 'Silver'

df['tier'] = df.apply(tier, axis=1)
print(df['tier'].value_counts().to_dict())
```

Q11 Predict the output · Medium

What does the first `print` in Exhibit 2B show?

A. `['Pune', 'Pune', 'Indore']`

B. `['Pune', 'Pune', 'Indore', 'Surat']`

C. `['Pune', 'Surat', 'Pune', 'Indore']`

D. Nothing: comparing NaN with 400 raises an error

Q12 Predict the output · Hard

What does the last `print` in Exhibit 2B show?

A. `{'Gold': 2, 'Platinum': 1, 'Silver': 2}`

B. `{'Gold': 3, 'Silver': 2}`

C. `{'Gold': 3, 'Silver': 1}`

D. Nothing: `row['amount'] >= 500` raises `TypeError` on the missing amount

How many customers really ordered, and how much?

Kalpa's warehouse runs on PostgreSQL. Kavya queries the three tables below. Assume standard PostgreSQL behaviour.

Exhibit 3A customers

cust_id	name	city
1	Asha	Pune
2	Ravi	Surat
3	Neha	Pune
4	Imran	NULL

Exhibit 3B orders

order_id	cust_id	amount	status
101	1	500	delivered
102	1	300	cancelled
103	2	700	delivered
104	2	NULL	delivered
105	5	400	delivered
106	3	200	cancelled

Exhibit 3C payments

order_id	paid
101	300
101	200
103	700

Q13 Choose the reason · Easy

This query fails with `column "gross" does not exist`:

```
SELECT order_id, amount * 1.18 AS gross FROM orders WHERE gross > 500;
```

Why?

A. `1.18` is a decimal and `amount` is an integer, so the types clash.

B. WHERE is evaluated before SELECT, so the alias `gross` does not exist yet when the filter runs.

C. An alias must be written in double quotes before it can be used anywhere else in the query.

D. WHERE can compare a column only with another column, not with a number.

Q14 Predict the row count · Medium

How many rows does this return?

```
SELECT c.name, o.order_id FROM customers c LEFT JOIN orders o ON c.cust_id = o.cust_id;
```

A. 4

B. 5

C. 6

D. 7

Q15 Predict the counts · Hard

Query X:

```
SELECT c.name, COUNT(o.order_id) AS n FROM customers c LEFT JOIN orders o ON c.cust_id = o.cust_id AND o.status = 'delivered' GROUP BY c.name;
```

Query Y is the same, except the status condition moves to a WHERE clause:

```
... LEFT JOIN orders o ON c.cust_id = o.cust_id WHERE o.status = 'delivered' GROUP BY c.name;
```

How many rows do X and Y return?

A. X: 4, Y: 4

B. X: 4, Y: 2

C. X: 2, Y: 2

D. X: 4, Y: 3

Q16 Predict the output · Medium

`SELECT COUNT(*), COUNT(amount), COUNT(DISTINCT cust_id), AVG(amount) FROM orders;` returns which four values?

A. 6, 5, 4, 420

B. 6, 6, 4, 350

C. 6, 5, 3, 420

D. 6, 5, 4, 350

Q17 Choose the query · Medium

Anand wants every customer whose delivered revenue is at least 600. Which query returns exactly that?

A. `SELECT cust_id, SUM(amount) AS rev FROM orders WHERE status = 'delivered' GROUP BY cust_id HAVING SUM(amount) >= 600;`

B. `SELECT cust_id, SUM(amount) AS rev FROM orders WHERE SUM(amount) >= 600 GROUP BY cust_id;`

C. `SELECT cust_id, SUM(amount) AS rev FROM orders GROUP BY cust_id HAVING SUM(amount) >= 600 AND status = 'delivered';`

D. `SELECT cust_id, SUM(amount) AS rev FROM orders GROUP BY cust_id HAVING SUM(amount) >= 600;`

Q18 Write the number · Hard

What value of `order_value` does this query return for Asha?

```
SELECT c.name, SUM(o.amount) AS order_value FROM customers c JOIN orders o ON c.cust_id = o.cust_id LEFT JOIN payments p ON o.order_id = p.order_id WHERE c.name = 'Asha' GROUP BY c.name;
```

Answer: _____

Who are the top sellers, and which way is revenue moving?

Meera Raghavan, the CEO, wants the top sales reps in each region and a month-on-month revenue view. Kavya uses window functions on the tables below (PostgreSQL).

Exhibit 4A rep_sales

rep	region	revenue
Anu	North	900
Bala	North	700
Chetan	North	900
Divya	North	500
Esha	South	800
Farhan	South	800
Gita	South	600

Exhibit 4B monthly (revenue in Rs lakh)

month	revenue
2026-01	100
2026-02	120
2026-03	120
2026-04	90

Exhibit 4C daily

day	amount
1	100
2	50
2	30
3	20

Q19 Predict the output · Medium

On Exhibit 4A, Kavya computes `ROW_NUMBER()`, `RANK()` and `DENSE_RANK()`, each `OVER (PARTITION BY region ORDER BY revenue DESC)`. What are the three values on Divya's row, in that order?

A. 4, 4, 3
B. 4, 3, 3
C. 4, 4, 4
D. 3, 3, 2

Q20 Write the number · Hard

How many rows does this return?

```
WITH r AS (SELECT rep, region, revenue, DENSE_RANK() OVER (PARTITION BY region ORDER BY revenue DESC) AS dr FROM rep_sales) SELECT * FROM r WHERE dr <= 2;
```

Answer: _____

Q21 Choose the query · Medium

Meera wants the top-revenue rep(s) in each region, ties included. Which query runs and returns them?

A. `SELECT rep FROM rep_sales WHERE RANK() OVER (PARTITION BY region ORDER BY revenue DESC) = 1;`
B. `WITH r AS (SELECT rep, region, RANK() OVER (PARTITION BY region ORDER BY revenue DESC) AS rk FROM rep_sales) SELECT rep, region FROM r WHERE rk = 1;`
C. `SELECT rep, RANK() OVER (PARTITION BY region ORDER BY revenue DESC) AS rk FROM rep_sales GROUP BY rep, region, revenue HAVING rk = 1;`
D. `SELECT rep, RANK() OVER (PARTITION BY region ORDER BY revenue DESC) AS rk FROM rep_sales WHERE rk = 1;`

Q22 Predict the output · Medium

```
SELECT month, revenue - LAG(revenue) OVER (ORDER BY month) AS change FROM monthly ORDER BY month;
```

What is `change` for January to April, in order?

A. NULL, 20, 0, -30

B. 0, 20, 0, -30

C. 20, 0, -30, NULL

D. NULL, -20, 0, 30

Q23 Choose the statement · Hard

On Exhibit 4C:

```
SELECT day, amount, SUM(amount) OVER (ORDER BY day) AS run_range, SUM(amount) OVER (ORDER BY day
ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS run_rows FROM daily;
```

Which statement is true?

A. `run_range` and `run_rows` are identical on every row.

B. On both day-2 rows `run_range` is 180, while `run_rows` reaches 180 only on the second of the two day-2 rows.

C. `run_range` is 150 on one day-2 row and 180 on the other.

D. On day 3, `run_range` is 200 and `run_rows` is 180.

Is the change real, or the noise of a sample?

Kalpa wants to know its average basket and whether UPI autopay adoption has really risen. z values you may need: 1.645 (one-sided 5%, two-sided 10%), 1.96 (two-sided 5%), 2.576 (two-sided 1%).

Q24 Make the call · Easy

Kavya wants to estimate the average basket of all Kalpa orders placed in October, across the app, the website and the stores. Which sample should she use?

A. The first 500 orders placed through the app in October

B. 500 orders drawn at random from all of October's orders, across every channel

C. All October orders from the Pune flagship, since it has the most data

D. The 500 largest October orders, since they matter most to revenue

Exhibit 5A

> A random sample of 100 October orders has a mean basket of Rs 640. From last year's data the population standard deviation is known: Rs 150.

Q25 Write the number · Medium

What is the upper bound of the 95% confidence interval for the mean basket, in rupees, to one decimal place?

Answer: _____

Q26 True or false, with the reason · Medium

Statement: "There is a 95% chance that the true mean basket of all October orders lies inside the interval from Q25."

A. True, because the interval was built with z = 1.96, which covers 95% of the normal curve.

B. True, because 95% of October's orders have baskets inside that interval.

C. False, because the 95% describes the method: about 95% of intervals built this way capture the true mean, and this one either does or does not.

D. False, because the interval is about the sample mean, which is certainly inside it.

Q27 True or false · Medium

"To make the Q25 interval half as wide at the same confidence level, Kavya needs to double the sample to 200 orders."

T (true) or **F** (false)

Exhibit 5B

> Last year 20% of Kalpa app users paid by UPI autopay. Meera asks: "Has adoption risen above 20%?" Kavya takes a random sample of 400 app users; 100 of them use UPI autopay.

Q28 Write the number · Medium

What is the z statistic for the test of H0: p = 0.20, to two decimal places?

Answer: _____

Q29 Write the number · Hard

Keep the same test: H0: p = 0.20, n = 400, the tail that Meera's question calls for, α = 0.05. What is the smallest number of autopay users out of 400 that would lead Kavya to reject H0?

Answer: _____

Q30 Assertion and reason · Hard

Kavya tests a new checkout page against the old one (two-sided, $\alpha = 0.05$) and gets $p = 0.12$.
Assertion: Kavya should not announce that the new checkout has no effect on conversion.
Reason: A p-value of 0.12 means there is a 12% chance that the new checkout has no effect.

A. Both are true, and the reason explains the assertion.
B. Both are true, but the reason does not explain the assertion.
C. The assertion is true; the reason is false.
D. The assertion is false; the reason is true.
E. Both are false.

Q31 Name the error · Medium

Kalpa tests new packaging with H0: "the damage rate does not change". The test fails to reject H0, so Kalpa keeps the old packaging. In truth, the new packaging does reduce damage. Which error is this, and what does it cost?

A. Type I: Kalpa rolls out packaging that does nothing.
B. Type II: Kalpa keeps paying for damage that the new packaging would have prevented.
C. Type I: Kalpa keeps paying for damage that the new packaging would have prevented.
D. Type II: Kalpa rolls out packaging that does nothing.

What is the data actually telling us?

Before the quarterly review, Kavya explores October's order data and a store-level comparison.

Exhibit 6A

> October order values: mean Rs 1,450, median Rs 620, mode Rs 450. First quartile (Q1) Rs 400, third quartile (Q3) Rs 800.

Q32 Choose the statement · Easy

Which statement fits Exhibit 6A?

A. The distribution is left-skewed, and the mean is the best "typical order".

B. The distribution is right-skewed; the median is the better "typical order", because a few large orders pull the mean up.

C. The distribution is roughly symmetric, since all three are averages of the same orders.

D. The distribution is right-skewed, so Kavya should delete the largest orders and report the mean of the rest.

Q33 Write the number · Medium

Using the 1.5 × IQR rule on Exhibit 6A, above what order value (in rupees) is an order flagged as a possible outlier?

Answer: _____

Q34 Put the steps in order · Medium

Put Kavya's EDA steps in order, first to last. Write the letters.

A. Look at each column's distribution on its own (histograms, box plots, counts)

B. Confirm what one row means and which columns are the keys

C. Check dtypes, missing values and duplicates

D. Compare segments (store, channel, month)

E. Look at relationships between pairs of variables

F. Write the conclusions, with their caveats

Order: ____ , ____ , ____ , ____ , ____ , ____

Exhibit 6B October visits and orders

Store	App visits	App orders	Web visits	Web orders
P	100	80	20	5
Q	20	18	100	30

Q35* Choose every correct option · Hard

Conversion = orders ÷ visits. Which statements about Exhibit 6B are true? Mark every correct option.

A. Q converts better than P in both channels.

B. P's overall rate is higher mainly because most of P's visits come through the app, where both stores convert far better.

C. P's overall conversion is 52.5%, the average of its two channel rates.

D. The overall figures show that P's staff convert visitors better than Q's.

E. Q's overall conversion is 40%.

** Mark every correct option. Right only with every correct box and no other.*

Q36 Make the call · Medium

Across Kalpa's 12 stores, Pearson's r between weekly footfall and weekly revenue is 0.85. The mall flagship has by far the highest footfall and revenue, and both figures are verified as real. Without it, r on the other 11 stores is 0.20. What should Kavya report?

A. r = 0.85: footfall strongly predicts revenue across stores.

B. Both figures: the flagship alone creates most of the correlation, and among the other 11 stores the linear link is weak.

C. Delete the flagship as an outlier and report r = 0.20 only.

D. r = 0.85, and that raising footfall will raise revenue.

Answer sheet

Circle one letter, or write the answer. Starred items: circle every correct letter.

Q1	Predict the output	A B C D
Q2	Choose the line	A B C D
Q3	Predict the output	A B C D
Q4	Write the number	_____
Q5*	Choose every correct option	A B C D E
Q6	Predict the output	A B C D
Q7	Predict the output	A B C D
Q8	Predict the counts	A B C D
Q9	Choose the line	A B C D
Q10*	Choose every correct option	A B C D E
Q11	Predict the output	A B C D
Q12	Predict the output	A B C D
Q13	Choose the reason	A B C D
Q14	Predict the row count	A B C D
Q15	Predict the counts	A B C D
Q16	Predict the output	A B C D
Q17	Choose the query	A B C D
Q18	Write the number	_____

Q19	Predict the output	A B C D
Q20	Write the number	_____
Q21	Choose the query	A B C D
Q22	Predict the output	A B C D
Q23	Choose the statement	A B C D
Q24	Make the call	A B C D
Q25	Write the number	_____
Q26	True or false, with the reason	A B C D
Q27	True or false	T F
Q28	Write the number	_____
Q29	Write the number	_____
Q30	Assertion and reason	A B C D E
Q31	Name the error	A B C D
Q32	Choose the statement	A B C D
Q33	Write the number	_____
Q34	Put the steps in order	_ _ _ _ _ _
Q35*	Choose every correct option	A B C D E
Q36	Make the call	A B C D

Solutions and marking key

Attempt the paper before reading on.

Marking

- Score one for each item that matches the key exactly: every correct letter and no other on a starred item (Q5, Q10, Q35), the whole sequence on Q34, and the number on a written item.
- Written numbers: Q4 accepts 275; Q18 accepts 1300 or 1,300; Q20 accepts 6; Q25 accepts 669.4 (669 is wrong); Q28 accepts 2.5 or 2.50; Q29 accepts 94 only; Q33 accepts 1400 or 1,400.
- Every printed result was checked by running it: the pandas exhibits on pandas 3.0, the SQL exhibits on PostgreSQL 16.
- About 70% of the items test Week 2 topics, about 20% the Week 1 ideas they build on, and about 10% revisit unfixed mistake-log items (Q8, Q27, Q28, Q30).

The key

Q	Key	Format	Part	Level	Skill
1	**B**	Predict the output	1	Easy	size vs count vs nunique
2	**B**	Choose the line	1	Medium	transform vs agg
3	**B**	Predict the output	1	Medium	to_datetime with format and coerce
4	**275**	Write the number	1	Medium	Grain: average of averages
5	**A, C**	Choose every correct option	1	Hard	filter, as_index, NaN in aggregates
6	**B**	Predict the output	1	Hard	Date arithmetic and a fixed AS_OF
7	**B**	Predict the output	2	Medium	Row explosion from duplicate keys
8	**B**	Predict the counts	2	Hard	drop_duplicates: full row vs subset (mistake log W1 Q6)
9	**C**	Choose the line	2	Medium	merge validate=
10	**A, B**	Choose every correct option	2	Hard	Anti-join and indicator
11	**A**	Predict the output	2	Medium	Boolean masks with NaN
12	**B**	Predict the output	2	Hard	apply(axis=1) and first-match rule order
13	**B**	Choose the reason	3	Easy	Logical order of execution
14	**C**	Predict the row count	3	Medium	LEFT JOIN row count
15	**B**	Predict the counts	3	Hard	Filter in ON vs WHERE for a LEFT JOIN
16	**A**	Predict the output	3	Medium	COUNT variants and AVG with NULL
17	**A**	Choose the query	3	Medium	WHERE vs HAVING
18	**1300**	Write the number	3	Hard	Fan-out inflating SUM
19	**A**	Predict the output	4	Medium	ROW_NUMBER vs RANK vs DENSE_RANK
20	**6**	Write the number	4	Hard	Top-N per group with ties
21	**B**	Choose the query	4	Medium	Filtering on a window result
22	**A**	Predict the output	4	Medium	LAG for period-over-period
23	**B**	Choose the statement	4	Hard	Default window frame: RANGE vs ROWS
24	**B**	Make the call	5	Easy	Sampling frame and bias
25	**669.4**	Write the number	5	Medium	CI for a mean, σ known
26	**C**	True or false, with the reason	5	Medium	Interpreting a confidence interval
27	**F**	True or false	5	Medium	SE and sample size (mistake log W1 Q27)
28	**2.50**	Write the number	5	Medium	One-sample z for a proportion (mistake log W1 Q28)
29	**94**	Write the number	5	Hard	Rejection boundary for a count
30	**C**	Assertion and reason	5	Hard	p-value interpretation; fail to reject ≠ H0 true (mistake log W1 Q30)

Q	Key	Format	Part	Level	Skill
31	**B**	Name the error	5	Medium	Type I and Type II errors
32	**B**	Choose the statement	6	Easy	Skew and the typical value
33	**1400**	Write the number	6	Medium	IQR outlier fence
34	**B, C, A, E, D, F**	Put the steps in order	6	Medium	EDA workflow
35	**A, B, E**	Choose every correct option	6	Hard	Simpson's paradox / mix effects
36	**B**	Make the call	6	Medium	Pearson's r and influential points

Model solutions: why each answer holds

Q1 Key B Predict the output · Easy · size vs count vs nunique

Why it holds. Surat has three rows: 103, 104 and 105. `size` counts rows, so 3. `count` counts non-missing values, and 105's amount is NaN, so 2. `nunique` counts distinct customers, C3 and C4, so 2.

A. It treats `count` like `size`. `count` skips NaN.

C. It treats `nunique` as a row count and counts C3 twice.

D. It has `size` dropping the NaN row. `size` never drops anything.

What interviewers look for. size against count said as "rows against non-missing values", and the gap (3 − 2 = 1) read as a data-quality signal: one failed payment.

Follow-up. Surat's `amount.mean()` is 300. Over how many orders? **Listen for:** Two. `mean` skips NaN, so the failed order is invisible unless you report size − count beside it.

Q2 Key B Choose the line · Medium · transform vs agg

Why it holds. `transform` returns one value per original row with the original index (0 to 4): 800, 800, 400, 400, 400. Division then lines up row by row: 200/800, 600/800, 100/400, 100/400, 200/400.

A. `agg` returns 2 rows indexed by store name. Dividing aligns labels 0 to 4 against 'Pune' and 'Surat', nothing matches, and every share is NaN, with no error.

C. It divides by the grand total, Rs 1,200, which gives the share of company sales (0.17, 0.50, ...), not of the store.

D. The same alignment problem as A, upside down: all NaN.

What interviewers look for. That `agg` changes the grain (one row per group) while `transform` keeps it (one row per original row).

Follow-up. How would you get the same column without `transform`? **Listen for:** `agg` with `as_index=False`, merge back on `store` with `validate='many_to_one'`, then divide. It works, but it is a join you then have to check.

Q3 Key B Predict the output · Medium · to_datetime with format and coerce

Why it holds. `format='%d/%m/%Y'` reads day first: 3 April, 15 April, 1 May. 31 April does not exist, and `errors='coerce'` turns it into NaT, so `isna().sum()` is 1. The months present are April and May; `nunique` ignores NaT, so 2.

A. It misses that 31 April is not a date.

C. It reads '03/04' month first as 3 March; the format string fixes day first.

D. That is `errors='raise'`, the default. With `coerce` the bad date silently becomes NaT.

What interviewers look for. An explicit format (or `dayfirst=True`) for Indian dates, and a count of NaT after every parse.

Follow-up. How do you send the bad date back to the store? **Listen for:** Keep the raw string, select rows where the parsed date is NaT but the raw text is not empty, and put them in a rejects log.

Q4 Key 275 Write the number · Medium · Grain: average of averages

Why it holds. Pune: revenue 5,000 + 6,000 = 11,000 on 10 + 30 = 40 orders, and 11,000 ÷ 40 = 275. The Rs 350 is the mean of the daily averages (500 and 200), which gives the quiet 10-order day the same weight as the busy 30-order day.

Accept. 275 or 275.0

What interviewers look for. Recompute a ratio from the summed numerator and denominator at the grain you report; never average averages unless the weights are equal.

Follow-up. Surat comes out at 250 both ways. Why? **Listen for:** Both Surat days had 20 orders, so the weights are equal and the simple and weighted means agree.

Q5* **Key A, C** **Choose every correct option · Hard · filter, as_index, NaN in aggregates**

Why it holds. A: groups have 2 (Pune), 3 (Surat) and 1 (Indore) rows; `filter` keeps whole groups with at least 2 rows, so 2 + 3 = 5 rows. C: `as_index=False` keeps `store` as a column, one row per store: 3 rows.

B. pandas `sum` skips NaN: Surat is 300 + 300 = 600.

D. `transform` keeps the original length: 6 values, not 3.

E. `mean` skips NaN: Pune is 500 ÷ 1 = 500. The 250 comes from treating the missing amount as 0.

What interviewers look for. Which groupby outputs are reduced (`agg`, `sum`, `mean`), which keep the length (`transform`) and which keep a subset of rows (`filter`).

Follow-up. Should Pune's mean count the failed order as zero? **Listen for:** No: `fillna(0)` would give 250, but a failed payment is not a Rs 0 order. Report it separately as a failed-payment count.

Q6 **Key B** **Predict the output · Hard · Date arithmetic and a fixed AS_OF**

Why it holds. Days to 1 October: C1 30; C2 90 (3 Jul to 3 Aug is 31, to 3 Sep 31, to 1 Oct 28); C3 91; C4 108. Strictly more than 90 keeps C3 and C4: 2.

A. It miscounts the 31-day months and drops C3 or C4.

C. It treats `>` as `>=` and includes C2 at exactly 90.

D. AS_OF is a fixed timestamp, so the answer is the same on any day. It would drift only with `pd.Timestamp.today()`.

What interviewers look for. A fixed reference date for reproducible reports, and the boundary (> or >=) written into the definition of "lapsed".

Follow-up. How would you write the rule as a date rather than a day count? **Listen for:** `cust['last_order'] < AS_OF - pd.Timedelta(days=90)`, that is before 3 July, which keeps C3 and C4 and leaves C2 out.

Q7 **Key B** **Predict the output · Medium · Row explosion from duplicate keys**

Why it holds. C2 appears twice in the master, so orders 2 and 3 each match 2 rows: 1 + 2 + 2 + 1 = 6 rows. The left join keeps order 4 (C9) with city NaN. Amounts repeat: 100 + 200×2 + 300×2 + 400 = 1,500, Rs 500 more than the true 1,000.

A. One row per order holds only when the master has one row per customer.

C. 5 rows and 1,100 is the inner join, which drops order 4.

D. It counts the duplicated rows but not their duplicated amounts.

What interviewers look for. Row count and revenue checked before and after every join.

Follow-up. What one argument would have caught this? **Listen for:** `validate='many_to_one'` on the merge (Q9).

Q8 **Key B** **Predict the counts · Hard · drop_duplicates: full row vs subset (mistake log W1 Q6)**

Why it holds. `drop_duplicates()` compares every column, and 'Surat' differs from 'Surat ' (trailing space), so nothing is dropped: 4. `subset='cust_id'` keeps the first C2: 3. With one master row per customer, each order matches at most one row: 4.

A. The trailing space makes the two C2 rows different, so full-row dedup keeps both.

C. A left join onto a master with unique keys cannot add rows.

D. Both mistakes at once.

What interviewers look for. Dedup on the identity key, after normalising text such as `str.strip()`.

Follow-up. Which C2 row should survive? **Listen for:** Strip the city first; then the rows are identical and nothing is lost. If they truly conflict, set a rule (for example the latest update) and log it.

Q9 **Key C** **Choose the line · Medium · merge validate=**

Why it holds. Many orders share one customer, and the master should hold one row per customer: many-to-one. `many_to_one` checks that the right-hand keys are unique, so it passes on a clean master and raises `MergeError` on Exhibit 2A, where C2 repeats.

A. `one_to_one` also needs the order keys unique, so it fails even on clean data: C2 ordered twice.

B. `one_to_many` checks the left side, the wrong direction; it also fails on clean data.

D. `many_to_many` checks nothing, so it never stops the run.

What interviewers look for. The relationship said aloud before the merge, then enforced with `validate`.

Follow-up. What would `customers.merge(orders, validate=...)` need? **Listen for:** `'one_to_many'`: the same relationship seen from the customer side.

Q10* Key A, B Choose every correct option · Hard · Anti-join and indicator

Why it holds. A: C9 is not in the master, so `~isin` keeps order 4 only. B: `indicator=True` marks rows with no match as `left_only`: order 4 only.

C. An inner join returns the matched orders (1, 2, 2, 3, 3).

D. City is NaN for order 4, but also for order 1, because C1 is in the master with a missing city. A NaN in a data column cannot tell "no match" from "matched, value missing".

E. `right_only` is customer C3, who has no orders: the opposite anti-join.

What interviewers look for. An anti-join by `isin` or `indicator`, never by testing a data column for NaN.

Follow-up. What does `m['_merge'].value_counts()` show for the left join in B? **Listen for:** both 5, left_only 1, right_only 0.

Q11 Key A Predict the output · Medium · Boolean masks with NaN

Why it holds. `amount > 400`: True, False (NaN), True, True, False. `channel == 'web'`: only row 1 is True (None == 'web' is False), so the `~` gives True, False, True, True, True. Both: rows 0, 2 and 3: Pune, Pune, Indore.

B. Surat's 300 fails `> 400`.

C. It treats `NaN > 400` as True. Any comparison with NaN is False.

D. Comparisons with NaN return False; they do not raise.

What interviewers look for. NaN in comparisons is False, so `~(x == v)` lets missing values through; brackets around each condition.

Follow-up. How would you also keep rows with an unknown amount for review? **Listen for:** `df['amount'].isna() | (df['amount'] > 400)`.

Q12 Key B Predict the output · Hard · apply(axis=1) and first-match rule order

Why it holds. Rules run top to bottom and the first return wins. 1,200 meets `>= 500` first, so it is Gold and Platinum is unreachable. NaN fails both comparisons, so it falls to Silver, as does 300. Gold: 500, 1,200, 800 = 3; Silver: 2.

A. It assumes 1,200 reaches Platinum; put the most specific rule first.

C. `apply` runs on every row, including the one with a missing amount.

D. Comparisons with NaN return False; they do not raise.

What interviewers look for. Rule order from most specific to least, and an explicit branch for missing values.

Follow-up. Rewrite `tier` so the missing amount is 'Unknown' and 1,200 is Platinum. **Listen for:** Check `pd.isna(row['amount'])` first, then `>= 1000`, then `>= 500`, else Silver.

Q13 Key B Choose the reason · Easy · Logical order of execution

Why it holds. Logical order: FROM/JOIN, WHERE, GROUP BY, HAVING, SELECT, DISTINCT, ORDER BY, LIMIT. The alias is created in SELECT, after WHERE has run.

A. PostgreSQL multiplies an integer by a decimal without complaint.

C. Double quotes control case and special characters, not where an alias is visible.

D. WHERE compares columns with constants all the time.

What interviewers look for. The logical order recited, with the consequence: an alias works in ORDER BY but not in WHERE.

Follow-up. Fix the query two ways. **Listen for:** Repeat the expression (`WHERE amount * 1.18 > 500`), or compute `gross` in a CTE and filter outside it.

Q14 Key C Predict the row count · Medium · LEFT JOIN row count

Why it holds. Asha has 2 orders, Ravi 2, Neha 1, and Imran none, so he is kept once with NULLs: 2 + 2 + 1 + 1 = 6. Order 105 (customer 5) has no customer row, so a left join from customers drops it.

A. One row per customer ignores the one-to-many join.

B. 5 is the inner-join count, which drops Imran.

D. 7 includes order 105, which only a FULL JOIN keeps.

What interviewers look for. Which side a LEFT JOIN protects, and that one-to-many joins multiply rows.

Follow-up. How many rows would a FULL JOIN return? **Listen for:** 7: the 6 above plus order 105 with a NULL name.

Q15 Key B Predict the counts · Hard · Filter in ON vs WHERE for a LEFT JOIN

Why it holds. X: the status test sits in ON, so every customer is kept and only delivered orders attach: Asha 1, Ravi 2 (103, 104), Neha 0, Imran 0, which is 4 rows. Y: WHERE runs after the join and drops rows where `o.status` is NULL or 'cancelled', so Neha and Imran disappear and the LEFT JOIN behaves like an inner join: 2 rows.

A. It misses that WHERE removes the customers with no delivered order.

C. It misses that ON keeps every customer.

D. Neha's only order is cancelled, so she is dropped in Y too.

What interviewers look for. "A filter on the right-hand table goes in ON if you want to keep the left join."

Follow-up. In X, why is Neha's n 0 and not 1? **Listen for:** `COUNT(o.order_id)` counts non-NULL values; `COUNT(*)` would count her NULL-padded row and give 1.

Q16 Key A Predict the output · Medium · COUNT variants and AVG with NULL

Why it holds. COUNT(*) counts 6 rows; COUNT(amount) skips order 104's NULL, so 5; COUNT(DISTINCT cust_id) counts 1, 2, 3 and 5, so 4; AVG ignores NULL: (500 + 300 + 700 + 400 + 200) ÷ 5 = 2,100 ÷ 5 = 420.

B. It counts the NULL amount and divides 2,100 by 6.

C. It drops customer 5, but this query reads only `orders`, where customer 5 exists.

D. It treats the NULL as 0 in AVG: 2,100 ÷ 6 = 350.

What interviewers look for. COUNT(*) against COUNT(col), and that aggregates skip NULL.

Follow-up. How do you get the average with the missing amount counted as 0? **Listen for:** `AVG(COALESCE(amount, 0))` = 350, and you should say why you chose that treatment.

Q17 Key A Choose the query · Medium · WHERE vs HAVING

Why it holds. WHERE keeps delivered rows before grouping; HAVING keeps groups after. Customer 2 has 700 (the NULL is skipped) and passes; customer 1 has 500 delivered and customer 5 has 400. Result: customer 2 only.

B. Aggregates are not allowed in WHERE; PostgreSQL raises an error.

C. `status` is neither grouped nor aggregated, so HAVING cannot test it; error.

D. No status filter: customer 1's cancelled 300 counts, so customer 1 wrongly qualifies with 800.

What interviewers look for. Row filters in WHERE, group filters in HAVING.

Follow-up. Can the status condition legitimately live inside HAVING? **Listen for:** Yes, inside an aggregate: `HAVING SUM(amount) FILTER (WHERE status = 'delivered') >= 600`, or a SUM of a CASE.

Q18 Key 1300 Write the number · Hard · Fan-out inflating SUM

Why it holds. Order 101 (500) has two payment rows, so the join gives two rows each carrying amount 500. Order 102 (300) has no payment and is kept once by the LEFT JOIN. SUM = 500 + 500 + 300 = 1,300, against a true order value of 800.

Accept. 1300 or 1,300

What interviewers look for. Fan-out spotted before it reaches a SUM: aggregate the many side first.

Follow-up. Write the correct query. **Listen for:** A CTE `pay AS (SELECT order_id, SUM(paid) AS paid FROM payments GROUP BY order_id)`, joined to orders, so each order has at most one row.

Q19 Key A Predict the output · Medium · ROW_NUMBER vs RANK vs DENSE_RANK

Why it holds. North in revenue order: Anu 900, Chetan 900, Bala 700, Divya 500. ROW_NUMBER numbers 1 to 4. RANK skips after the tie: 1, 1, 3, 4. DENSE_RANK has no gaps: 1, 1, 2, 3. Divya: 4, 4, 3.

B. It has RANK not skipping after the tie.

C. It has DENSE_RANK skipping a number.

D. Those are Bala's values.

What interviewers look for. The three functions explained on a tie.

Follow-up. Which of Anu and Chetan gets ROW_NUMBER 1? **Listen for:** Either: it is arbitrary unless the ORDER BY has a tie-breaker, such as `revenue DESC, rep`.

Q20 Key 6 Write the number · Hard · Top-N per group with ties

Why it holds. DENSE_RANK in North: Anu 1, Chetan 1, Bala 2, Divya 3, so 3 rows have dr ≤ 2. South: Esha 1, Farhan 1, Gita 2, so 3 rows. Total 6. With RANK, Bala and Gita would rank 3 and the count would be 4; ROW_NUMBER also gives 4.

What interviewers look for. Which ranking matches the business rule: the top two revenue levels (DENSE_RANK) or the top two people (ROW_NUMBER).

Follow-up. Meera wants exactly two reps per region. How? **Listen for:** ROW_NUMBER with a stated tie-breaker, and a note telling her who was left out by the tie.

Q21 Key B Choose the query · Medium · Filtering on a window result

Why it holds. Window functions are computed with SELECT, after WHERE and HAVING. To filter on a rank, compute it in a CTE (or subquery) and filter outside. B returns Anu, Chetan, Esha and Farhan.

A. Window functions are not allowed in WHERE; error.

C. The alias `rk` does not exist when HAVING runs; error.

D. The alias is not visible in WHERE; error.

What interviewers look for. That a window result is filtered one level up.

Follow-up. Other than a CTE, how can you do it? **Listen for:** A subquery in FROM: `SELECT ... FROM (SELECT ..., RANK() OVER (...) AS rk FROM rep_sales) t WHERE rk = 1`. PostgreSQL has no QUALIFY.

Q22 Key A Predict the output · Medium · LAG for period-over-period

Why it holds. LAG fetches the previous row's revenue. January has none, so 100 − NULL is NULL. Then 120 − 100 = 20, 120 − 120 = 0, 90 − 120 = −30. The text months 'YYYY-MM' sort correctly.

B. January has no previous month; the result is NULL, not 0.

C. That is LEAD(revenue) − revenue: next month minus this month.

D. The subtraction is reversed (previous minus current).

What interviewers look for. LAG and LEAD placed correctly, and the NULL in the first period explained.

Follow-up. Give the percentage change instead. **Listen for:** `(revenue - LAG(revenue) OVER w) * 100.0 / LAG(revenue) OVER w`: February 20%, March 0%, April −25%.

Q23 Key B Choose the statement · Hard · Default window frame: RANGE vs ROWS

Why it holds. With ORDER BY and no frame, the default is RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW, which includes every peer with the same day. So both day-2 rows show 100 + 50 + 30 = 180. ROWS counts physical rows: the first day-2 row shows 150 or 130 (the tie order is arbitrary) and the second shows 180. Day 3 is 200 in both.

A. They agree only when there are no ties in the ORDER BY.

C. That is the ROWS behaviour, not RANGE.

D. Both columns are 200 on day 3.

What interviewers look for. Knowing the default frame, and that running totals over tied keys need ROWS plus a tie-breaker.

Follow-up. How do you make the running total deterministic? **Listen for:** Add a unique tie-breaker to the ORDER BY (for example `day, order_id`) and use ROWS.

Q24 Key B Make the call · Easy · Sampling frame and bias

Why it holds. The population is every October order in every channel, so the sample must be drawn at random from all of them.

A. App-only orders may have different baskets: sampling bias.

C. One store is not the population, and it may be atypical.

D. The largest orders bias the estimate upward by design.

What interviewers look for. The sampling frame matched to the question before any formula.

Follow-up. Would a bigger app-only sample fix the problem? **Listen for:** No: more of a biased sample is still biased; it only gets more confidently wrong.

Q25 Key 669.4 Write the number · Medium · CI for a mean, σ known

Why it holds. SE = 150 ÷ √100 = 150 ÷ 10 = 15. Margin = 1.96 × 15 = 29.4. Upper bound = 640 + 29.4 = 669.4 (the interval is 610.6 to 669.4).

Accept. 669.4 or 669.40; 669 is wrong

What interviewers look for. SE said before the margin: the spread of the average, not of single orders.

Follow-up. What is the 99% interval, and why is it wider? **Listen for:** 2.576 × 15 = 38.64, so 601.36 to 678.64: more confidence costs width.

Q26 Key C True or false, with the reason · Medium · Interpreting a confidence interval

Why it holds. μ is fixed; the interval is what varies from sample to sample. 95% is the long-run share of intervals built this way that capture μ. Any one interval either contains μ or does not.

A. z = 1.96 sets the method's coverage; it does not turn a fixed interval into a probability statement about μ.

B. The interval is for the mean, not for individual orders, which spread with σ = 150.

D. The sample mean sits at the centre by construction; the interval is a statement about μ.

What interviewers look for. The method-not-interval wording, then a plain sentence for a manager.

Follow-up. How would you say it to Meera? **Listen for:** "Our best estimate is Rs 640; the true average is very likely between about Rs 611 and Rs 669. The method we used gets it right 95 times in 100."

Q27 Key F True or false · Medium · SE and sample size (mistake log W1 Q27)

Why it holds. The width is proportional to 1 ÷ √n. Doubling n divides the width by √2 (about 29% narrower). Halving it needs four times the sample: 400 orders.

What interviewers look for. The square root inside the SE, and its cost: four times the data for half the width.

Follow-up. What is the margin at n = 400? **Listen for:** SE = 150 ÷ 20 = 7.5, margin = 1.96 × 7.5 = 14.7, exactly half of 29.4.

Q28 Key 2.50 Write the number · Medium · One-sample z for a proportion (mistake log W1 Q28)

Why it holds. Under H0 the SE uses p0: √(0.20 × 0.80 ÷ 400) = √0.0004 = 0.02. $\hat{p}$ = 100 ÷ 400 = 0.25. z = (0.25 − 0.20) ÷ 0.02 = 2.50. Conditions: n·p0 = 80 and n(1 − p0) = 320, both ≥ 10.

Accept. 2.5 or 2.50

What interviewers look for. p0 in the SE for a test ($\hat{p}$ is for an interval), and the conditions checked.

Follow-up. What is the decision at α = 0.05, right-tailed? And at α = 0.01, two-sided? **Listen for:** 2.50 > 1.645, so reject. 2.50 < 2.576, so fail to reject: this is why the tail and α are fixed before looking at the data.

Q29 Key 94 Write the number · Hard · Rejection boundary for a count

Why it holds. Meera asks "risen above", so the test is right-tailed: reject when z > 1.645, that is $\hat{p}$ > 0.20 + 1.645 × 0.02 = 0.2329. As a count: 0.2329 × 400 = 93.16, so the smallest whole count is 94. Check: 94 ÷ 400 = 0.235 gives z = 1.75 (reject); 93 ÷ 400 = 0.2325 gives z = 1.625 (do not reject).

Accept. 94 only

What interviewers look for. The tail read from the question, and the boundary turned into a number the business can watch.

Follow-up. With n = 1,600, where is the boundary as a proportion? **Listen for:** SE = 0.01, so 0.20 + 0.01645 = 0.21645: the same deviation becomes significant with a larger n.

Q30 Key C Assertion and reason · Hard · p-value interpretation; fail to reject ≠ H0 true (mistake log W1 Q30)

Why it holds. The assertion is true: p = 0.12 > 0.05 means the data are not strong enough to reject H0; it does not prove there is no effect (the test may be too small). The reason is false: a p-value is the chance of data at least this extreme if H0 were true, not the chance that H0 is true.

A. The reason is false, so it cannot explain anything.

B. The reason is false, so it is not a true statement either.

D. The assertion is true.

E. The assertion is true.

What interviewers look for. "Fail to reject" said in plain words, and the p-value defined as conditional on H0.

Follow-up. Kavya's manager suggests switching to α = 0.15 so the result counts. Your reply? **Listen for:** No: α is fixed before the data are seen. Report the observed lift with its interval, and size a larger test for the smallest lift that matters.

Q31 Key B Name the error · Medium · Type I and Type II errors

Why it holds. H0 (no change) is false, yet the test kept it: a Type II error, a missed effect. Its cost is continuing to pay for damage the better packaging would have prevented.

A. A Type I error rejects a true H0; here H0 was kept.

C. The cost is right but the label is wrong.

D. The label is right but the cost belongs to a Type I error.

What interviewers look for. Both errors named in business terms: a false alarm versus a missed effect, each with its cost.

Follow-up. How could Kalpa make this error less likely? **Listen for:** A larger sample (more power), or a looser α if false alarms are cheap.

Q32 Key B Choose the statement · Easy · Skew and the typical value

Why it holds. Mode 450 < median 620 < mean 1,450: a long right tail of large orders pulls the mean up. The median describes the typical order.

A. A left skew would put the mean below the median.

C. A symmetric shape has mean and median close together; here the mean is more than double the median.

D. Large orders are real revenue: flag them, never delete them.

What interviewers look for. The median as the typical value for skewed money data, with the IQR as its spread.

Follow-up. Which spread goes beside the median? **Listen for:** The interquartile range, Rs 400 to Rs 800.

Q33 Key 1400 Write the number · Medium · IQR outlier fence

Why it holds. IQR = 800 − 400 = 400. 1.5 × 400 = 600. Upper fence = 800 + 600 = 1,400. (The lower fence, 400 − 600 = −200, flags nothing.)

Accept. 1400 or 1,400

What interviewers look for. The fence as a flag for review, not a deletion rule.

Follow-up. What do you do with the orders above Rs 1,400? **Listen for:** Check whether each is an error or real; keep the real ones, and report the figures with and without them.

Q34 Key B, C, A, E, D, F Put the steps in order · Medium · EDA workflow

Why it holds. Grain and keys (B), then data quality (C), then one variable at a time (A), then pairs (E), then segments (D), then conclusions with caveats (F). Each step relies on the one before: a segment comparison on duplicated rows explains nothing.

Accept. B, C, A, E, D, F only

What interviewers look for. The order more than any single step: understand, check, describe, relate, compare, conclude.

Follow-up. You find 3% duplicate orders at step C. What changes downstream? **Listen for:** Every count and total after it: dedup on the identity key, log what was removed, and rerun.

Q35* **Key A, B, E** **Choose every correct option · Hard · Simpson's paradox / mix effects**

Why it holds. Channel rates: P app 80/100 = 80%, web 5/20 = 25%; Q app 18/20 = 90%, web 30/100 = 30%. Q wins in both channels (A). Overall: P 85/120 ≈ 70.8%, Q 48/120 = 40% (E). P wins overall because 100 of its 120 visits are app visits (B).

C. Averaging rates ignores the denominators: (80 + 25) ÷ 2 = 52.5%, but the true rate is 85 ÷ 120 ≈ 70.8%.

D. The overall gap is a channel-mix effect; within each channel Q converts better.

What interviewers look for. Segment rates checked before an overall comparison, and the mix named as the cause.

Follow-up. How would you compare the stores fairly? **Listen for:** Compare within channel, or apply both stores' channel rates to the same channel mix.

Q36 **Key B** **Make the call · Medium · Pearson's r and influential points**

Why it holds. Pearson's r is pulled hard by one extreme point. The flagship is real, so it stays in the data, but the honest finding is that among typical stores the linear link is weak. Report both.

A. One store creates most of the 0.85, so it overstates the relationship for the other 11.

C. Deleting a verified real store hides part of the business.

D. Correlation is not causation: mall location could drive both footfall and revenue.

What interviewers look for. r reported with and without the influential point, a scatter plot, and correlation kept apart from cause.

Follow-up. What would you look at next? **Listen for:** The scatter plot, Spearman's rank correlation, and a confounder such as location type.